MOHAMMAD KAMRUL HASAN

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Education

Bangladesh University of Engineering and Technology (BUET) **Apr 2025 – Present**
B.Sc. in Computer Science and Engineering *Dhaka, Bangladesh*

Experience

Backend Developer Intern **Jun 2026 – Present**
Version3.ai *Dhaka, Bangladesh*

- Built the multi-tenant product database service in **NestJS**, **MongoDB** and **RabbitMQ**, keeping every store's data isolated so many merchants can share one platform safely.
- Built one-click store connection for **2** e-commerce platforms (WooCommerce, Shopify) with **OAuth 2.0** and encrypted credentials, so merchants onboard without handling API keys by hand.
- Evaluated image-search models and vector-database settings for visual product search, showing it can run on CPU-only servers instead of GPUs, with **2.2**× more searches per server.

Software Engineering Intern **Dec 2025 – Mar 2026**
Edulytics *Dhaka, Bangladesh*

- Built mobile screens in **React Native** and full-stack features in **Next.js** for an education platform's MVP.

Projects

LITTLETai: Faster, Lower-Energy LLM Inference on Android [GitHub] Arm AI Optimization Challenge
llama.cpp • Python • Shell • simpleperf • Perfetto *2026*

- Made **llama.cpp** generate text **2.2–2.6**× **faster** with **2.6–3.3**× **less energy** on a budget Android phone by choosing the right CPU cores and thread counts.
- Used **simpleperf** to trace the slowdown to threads spin-waiting, and found the core-pinning flag was ignored on Android.
- Built a reproducible benchmark harness (thermal cooldowns, energy logging, **209** tests) so every result can be regenerated.

NodeCommerce: Demand Forecasting for Bangladesh [GitHub] AI Buildfest hackathon
Python • FastAPI • LightGBM • SHAP • PostgreSQL *2026*

- Forecast stock demand for **495** upazilas across all **64** districts with **LightGBM**, with SHAP explaining each forecast.
- Automated **8** scheduled pipeline jobs (features, inference, spike detection, retraining) behind a **FastAPI** service.
- Built a map dashboard of regional surplus and shortage, so suppliers can see where stock is needed.

DroneCam: Tiny-Object Detection in Drone Footage [GitHub]
Python • Ultralytics YOLO • SAHI *2026*

- Reached **0.728 mAP@0.5** on VisDrone by training at high resolution and tiling frames, so distant objects aren't lost.
- Offered two tracking modes: an accurate one for analysis and a fast one for real time.

Ilham: Hadith Study Platform [GitHub]
PostgreSQL • Node.js • Express • TypeScript *2026*

- Designed the **PostgreSQL** database and ETL for **14,941** hadiths and their narrator chains, matching **99.6%** of narrators.
- Scored each hadith's reliability in SQL from the weakest narrator in its chain.
- Locked the reference data read-only at the database level, so the app can never corrupt it.

Sentinel: Offline Medical Triage App [GitHub]
React Native • TypeScript • TensorFlow Lite *2025*

- Led a team of **3** building an Android app that triages symptoms as Critical, Urgent or Routine fully offline.
- Paired a small on-device classifier with a **20 MB** language-model fallback for unclear inputs.
- Added voice input, local history and one-tap emergency calling; released as an APK.

Bayyinah: Collaborative Desktop Quran App [GitHub]
Java • JavaFX • Spring Boot • WebRTC • SQLite *2026*

- Built instant offline full-text search over all **6,236** verses in a **JavaFX** desktop client.
- Added live group reading with real-time **WebRTC** audio on a **Spring Boot** server, plus offline bookmark sync.

Publications

M. Rahaman, S. S. Rahman, M. S. M. Noor, K. S. S. Siddiquee, **M. K. Hasan**, et al., "From Wearable Data to Risk Literacy: Visualizing Physiological Heat Stress for Rickshaw Pullers in Dhaka," in *Proc. UbiComp/ISWC 2026, ISWC Notes & Briefs*, ACM, 2026. doi:10.1145/3830727.3834827

Skills

Languages: C, C++, Python, TypeScript, JavaScript, Java, SQL, POSIX shell
ML & inference: llama.cpp, PyTorch, TensorFlow Lite, Keras, Hugging Face, sentence-transformers, FAISS, Qdrant, OpenCV
Backend & data: NestJS, FastAPI, Spring Boot, Express, WebSockets, WebRTC, PostgreSQL, MongoDB, RabbitMQ, SQLite
Embedded & tools: ESP32, Linux, Git, GitHub Actions, simpleperf, Perfetto, adb, Ghidra